Let $\sigma \in \mathrm{Hom}(V, W)$ Then σ is injective $\Leftrightarrow \exists \tau \in \mathrm{Hom}(W, V)$ s.t. $\tau\sigma = I_V$

$$V \longrightarrow W = \mathrm{Im}\,\sigma \oplus W'$$

$$v \overset{\sigma}{\longmapsto} \sigma(v) \in \mathrm{Im}\,\sigma$$

$$v = \tau(w) \overset{\tau}{\longleftarrow} W = w \overset{\sigma(v)}{\underset{}{=}} + w'$$

<u>Recall</u> CHV Eigen-Theory

<u>Thm</u> Let $\dim V = n < +\infty$ $\sigma \in \mathrm{End}\,V$. Then the eigen-theory of σ is completely the same as <u>any matrix</u> associated to σ.

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<u>cor</u> Every theory of σ is almost the same as any matrix associated to σ

Ex. $\operatorname{rank}\sigma = \dim \mathrm{Im}\,\sigma$; $\operatorname{nullity}\sigma = n - \operatorname{rank}\sigma$; $\det\sigma = \prod_{i=1}^{n} \lambda_i$

σ is diagonalizable $\Leftrightarrow V = \sum \oplus V_\lambda$

<u>restriction of σ on subspaces</u>

$\sigma|_{V_\lambda}: \quad V_\lambda \longrightarrow V_\lambda$ $\qquad \forall v \in V,\ \sigma(v) = (\sum \oplus \sigma|_{V_\lambda})(\sum v_\lambda)$

$\qquad v \longmapsto \sigma|_{V_\lambda}(v) = \sigma(v) = \lambda v$ $\qquad = \sum \lambda v_\lambda$

(eigen vectors)

$\lambda \neq \mu$ two E.V. $\quad V_\lambda \oplus V_\mu \longrightarrow V_\lambda \oplus V_\mu$

Inducing $\tilde{\sigma} \in \mathrm{End}\,U$, $U \in \mathcal{L}(V)$ $\quad \tilde{\sigma}: U \to U$ $\qquad$ [U is an invariant subspace]

$$u \mapsto \tilde{\sigma}(u) = \sigma(u)$$

<u>Def</u> Let $\sigma \in \mathrm{End}\,V$ then $U \in \mathcal{L}(V)$ is called to be an "<u>invariant subspace</u>" of σ if $\sigma(U) \subseteq U$ (i.e. $\forall u \in U$, $\sigma(u) \in U$)

<u>Thm</u> Let $\sigma \in \mathrm{End}\,V$ Suppose U is an invariant subspace. Then $U = \sum F\alpha_i$ α_i is E.V. of σ.

<u>Application</u>

- $AB = BA \Leftrightarrow A$ and B have the common eigen vectors.

pf. $F^n \xrightarrow{A} F^n \xrightarrow{B} F^n$

$$x \longmapsto \overset{(y)}{Ax} \longmapsto By$$

$\forall V_\lambda$ an Eigen-subspace of B is inv under A, namely $A(V_\lambda) \subseteq V_\lambda$

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$A|_{V_\lambda}: V_\lambda \to V_\lambda$

$\quad v \mapsto Av$

Lack of Geometry $V/\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$

Def. Let $(\cdot,\cdot): V\times V \to \mathbb{F}$ satisfy

i) positivity: $(v,v) \geq 0$ and "=" $\iff v=0$

ii) symmetry $(u,v) = (v,u)$

iii) bilinear (双线性) $(au+a'u', v) = a(u,v) + a'(u',v)$

$(u, xv+x'v') = \bar{x}(u,v) + \bar{x}'(u,v')$

Then $(\cdot,\cdot)$ is an inner product on V, and $(V,(\cdot,\cdot))$ is an IPS (=Inner product space)

Ex. ① $V=\mathbb{R}$, $(x,y) = axy$ $(a>0)$

② $V=\mathbb{R}^2$, $\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}, \begin{pmatrix}y_1\\y_2\end{pmatrix}\right) = ax_1y_1 + cx_2y_2 + b(x_1y_2+x_2y_1)$ $\quad \begin{pmatrix}a & b\\ b & c\end{pmatrix} > 0$

③ $V = \mathbb{R}[x]_2$ $\quad (f(x), g(x))$

$\cong \mathbb{R}^2$ $\begin{pmatrix}1\\x\end{pmatrix}$

Big Thm Let $\dim V = n$. Then the inner product structure are essentially the same as $\mathbb{F}^n$. Namely, fix a basis of V $(v_1,\dots,v_n)$. Let $(\cdot,\cdot)$ be an IP on V. Then $(X,Y) = x^T A y$, where $A = ((v_i,v_j))$ (is necessarily positive def).

Pf. $X = (v_1,\dots,v_n)x$ $\quad Y = (v_1,\dots,v_n)y$

$(X,Y) = (\sum x_i v_i, \sum y_j v_j) = \sum_{i,j=1}^n x_i y_j (v_i,v_j) = x^T A y$

Thm IPs on $\mathbb{F}^n$ are bijectively correspond to positive def matrices.